

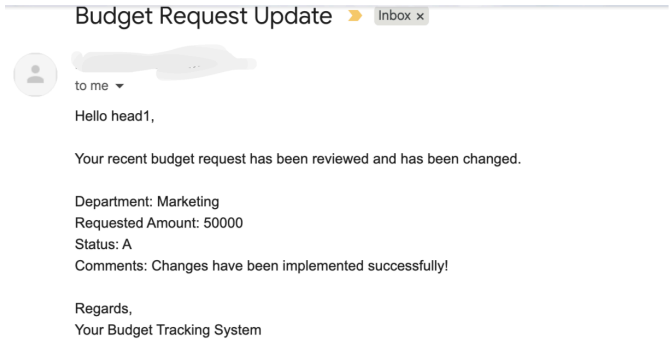
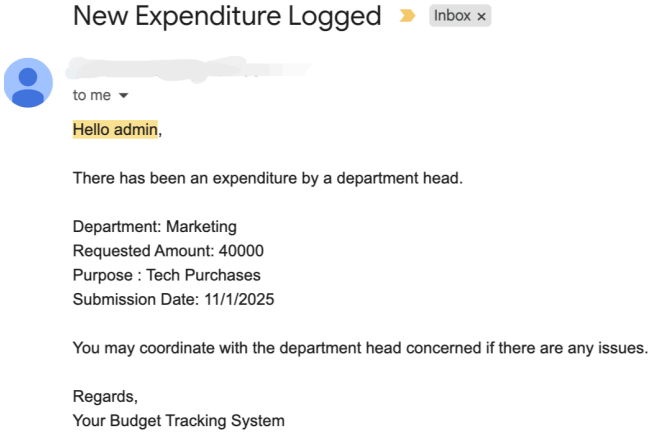
Computer Science Internal Assessment Criterion E - Evaluation

Refer to Appendix B for detailed client feedback.
Full Word Count: 508

No.	Criterion for Success	Met?	Evaluation																																				
1	Authentication features with role-based access (Admin, Department Head, Employee).	YES	RBAC via Express sessions worked well; client confirmed clear data separation and secure access.																																				
2	Secure session management with auto timeout.	YES	Sessions persist during activity and expire after inactivity, improving usability and protection.																																				
3	All passwords hashed.	YES	Bcrypt ensures secure hashing before storage; client confident about data safety. <table><tr><td>1</td><td>user1</td><td>newUserPass123</td><td>General Employee</td></tr><tr><td>2</td><td>admin</td><td>\$2b\$12\$a/K07LpINZGi3M.JdrHy0u.lunyhx3A.g...</td><td>admin</td></tr><tr><td>3</td><td>head123</td><td>\$2b\$10\$/gXBsgZJJUOhKfhJQ.ELbuxXRRGAH...</td><td>Department Head</td></tr><tr><td>6</td><td>testadmin234</td><td>\$2b\$10\$/pjvdaETXOicJqlqr9oTMOTZLedyD47z...</td><td>admin</td></tr><tr><td>7</td><td>user123</td><td>\$2b\$10\$.E712QF.8oMWES73ogEXOejlGQ2M....</td><td>General Employee</td></tr><tr><td>8</td><td>head3</td><td>\$2b\$10\$/tll/8GHKYOm571IbbVO3AO9uwgAGoi...</td><td>Department Head</td></tr><tr><td>9</td><td>head1</td><td>\$2b\$10\$/MijOGkO6hboGcG47sM2oeON3.f5Utg...</td><td>Department Head</td></tr><tr><td>10</td><td>head22</td><td>\$2b\$10\$/u5NhlU2V63aISbHta54oYOykUxpY/8a...</td><td>Department Head</td></tr><tr><td>12</td><td>user2</td><td>\$2b\$10\$/F7bkEggpe1XAOm2w.p3dONk4P1Gw...</td><td>General Employee</td></tr></table>	1	user1	newUserPass123	General Employee	2	admin	\$2b\$12\$a/K07LpINZGi3M.JdrHy0u.lunyhx3A.g...	admin	3	head123	\$2b\$10\$/gXBsgZJJUOhKfhJQ.ELbuxXRRGAH...	Department Head	6	testadmin234	\$2b\$10\$/pjvdaETXOicJqlqr9oTMOTZLedyD47z...	admin	7	user123	\$2b\$10\$.E712QF.8oMWES73ogEXOejlGQ2M....	General Employee	8	head3	\$2b\$10\$/tll/8GHKYOm571IbbVO3AO9uwgAGoi...	Department Head	9	head1	\$2b\$10\$/MijOGkO6hboGcG47sM2oeON3.f5Utg...	Department Head	10	head22	\$2b\$10\$/u5NhlU2V63aISbHta54oYOykUxpY/8a...	Department Head	12	user2	\$2b\$10\$/F7bkEggpe1XAOm2w.p3dONk4P1Gw...	General Employee
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4	Users can log in and reset passwords.	YES	Password reset emails sent instantly; client praised reliability. Forgot Password? Enter Your Username: admin Send Verification Code Enter Verification Code Enter the 5-digit code sent to your email: Verify Code																																				

			Enter new status for department 1: Use A / D / P or Approved / Denied / Pending Denied Cancel OK Status codes: A = Approved, D = Denied, P = Pending budget requests and approvals <table><tr><th>Department</th><th>Department Head</th><th>Amount</th><th>Approve/Deny</th><th>Comments</th></tr><tr><td>Marketing</td><td>head1</td><td>₹50000</td><td> D Edit status? </td><td> Changes have been implemented successfully! Edit comment? </td></tr><tr><td>Finance</td><td>head123</td><td>₹76000</td><td> A Edit status? </td><td> Changes have been incorporated; budget approved! Edit comment? </td></tr></table>	Department	Department Head	Amount	Approve/Deny	Comments	Marketing	head1	₹50000	D Edit status?	Changes have been implemented successfully! Edit comment?	Finance	head123	₹76000	A Edit status?	Changes have been incorporated; budget approved! Edit comment?										
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10	Department Heads can edit their department's budgets only.	YES	Proper validation prevents cross-department access.																									
11	Department Heads can submit expenditure requests.	YES	Requests automatically notify admin; client noted smoother workflow. Department Expenditures SPENDING TRACKER <table><tr><th>Serial No.</th><th>Expenditure Purpose</th><th>Expenditure Amount</th><th>Date</th><th>Delete</th></tr><tr><td>1</td><td>Tech Purchases</td><td>40,000</td><td>1/11/2025</td><td> Delete </td></tr><tr><td>2</td><td>Testing</td><td>50,000</td><td>25/9/2025</td><td> Delete </td></tr><tr><td>3</td><td>Employee Outing</td><td>45,000</td><td>14/8/2025</td><td> Delete </td></tr><tr><td>4</td><td>Advertising Campaign</td><td>15,000</td><td>1/5/2025</td><td> Delete </td></tr></table> Add Expenditure Add Expenditure Expenditure Purpose: Client Dinner Expenditure Amount: 2000 Date: 08/01/2026 Save Cancel	Serial No.	Expenditure Purpose	Expenditure Amount	Date	Delete	1	Tech Purchases	40,000	1/11/2025	Delete	2	Testing	50,000	25/9/2025	Delete	3	Employee Outing	45,000	14/8/2025	Delete	4	Advertising Campaign	15,000	1/5/2025	Delete
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12	Department Heads can track approvals.	YES	Status changes updated instantly with clear progress display. Department Head Dashboard BUDGET APPROVAL STATUS <table><tr><th>Department Head</th><th>Amount</th><th>Approve/Deny</th><th>Comments</th></tr><tr><td>head1</td><td>₹50,000</td><td>D</td><td>Changes have been i</td></tr></table>	Department Head	Amount	Approve/Deny	Comments	head1	₹50,000	D	Changes have been i																	
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13	Admin can review/comment on adjustments.	YES	Feedback stored in database; client valued this addition for clarity.																									

			Status codes: A = Approved, D = Denied, P = Pending
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19	New users receive onboarding guide.	YES	Client reported smooth onboarding and reduced training time.
20	Custom dashboards per role.	YES	Distinct layouts enhanced clarity; client approved separation.
21	General Employees and Department Heads should receive an email notification whenever a budget is approved or rejected by admin.	YES	Admin should receive an email notification whenever a budget request is submitted and pending approval. 
22	Admin should receive an email notification whenever an expenditure's been made by a department-head.	YES	Emails triggered instantly upon submission; client found this crucial for staying informed. 

Evaluation Word Count : 378

Recommendations

1. Predictive Budget Analytics (AI-driven insights).

Implement a machine-learning module (e.g., using Python + scikit-learn) to forecast future spending trends. Through Node.js *child processes*, Python scripts can run asynchronously alongside the existing Express server. The algorithm would train on historical **expenditures** and **approved_budget** data from MySQL, outputting predictions or alerts when projected expenses exceed limits.¹ This directly supports Mrs. Usha's requirement for proactive, data-driven HR budgeting.

2. Multi-format, data-rich reports with interactive analytics.

Extend report generation beyond jsPDF by integrating **exceljs** (for .xlsx) and **docx** libraries.² Add an export-type dropdown in the Admin dashboard and embed interactive Chart.js visualizations directly inside reports. MySQL queries can aggregate by **MONTH()** or **DEPARTMENT_ID** to generate comparative sheets for detailed quarterly reviews. This aligns with the client's feedback for enhanced visual and editable reports.³

Recommendation word count : 130

¹ Acquah, Emmanuel. "Machine Learning Models for Accurate Project Budget Forecasting." *Dartai.com*, Dart, 27 July 2025, www.dartai.com/blog/machine-learning-models-for-accurate-project-budget-forecasting? Accessed 31 Oct. 2025.

² Wai Foong Ng. "Guide to ExcelJS: Excel Workbook Manager | Built In." *Built In*, 2022, builtin.com/software-engineering-perspectives/exceljs? Accessed 31 Oct. 2025.

³ Zane. "Building a High-Quality Stock Report Generator with Node.js, Express, and OpenAI API." *DEV Community*, 3 Nov. 2024, dev.to/zand/building-a-high-quality-stock-report-generator-with-nodejs-express-and-openai-api-3ipm. Accessed 31 Oct. 2025.

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